SYSTEMATIC PORTFOLIOS / INDEPENDENT RESEARCH

Trade less, rebalance selectively

Threshold triggers and partial corrections

QuantLab Research

20 October 2026

POLICY GRID
410
Trigger bands and correction fractions

TRAINING-SELECTED BAND
3.75 pp
Correct 20% of the observed deviation

HOLDOUT TRADE REDUCTION
62.1%
Relative to weekly 50/50

Less trading is achievable; equal performance is not guaranteed. A simple threshold rule cuts holdout fills by about 62%, but its holdout CAGR trails weekly 50/50 by 1.38 percentage points. This is an implementation tradeoff, not a free improvement.

Executive findings

This study keeps the AAPL/TSLA target at 50/50 and replaces fixed-calendar trading with two decisions: how far the observed weight must move before an order is triggered, and what fraction of that deviation to correct. The 2020–2025 grid contains 41 bands and ten correction fractions, plus five calendar / buy-and-hold references.

A turnover-minimizing selection rule is fixed before evaluating the calculations. Using only 2020–2022, it selects a 3.75-percentage-point trigger and a 20% correction. A weight of 54%, for example, produces a target of 53.2%, not an immediate return to 50%. Execution occurs at the next open.

Across the full sample, this candidate makes 269 fills versus 613 for weekly 50/50. Its CAGR is 46.45% versus 46.61%, with a similar worst closing drawdown of approximately 55%. These full-sample results are descriptive because they include the training observations.

On a fresh 2023–2025 deployment, the selected rule makes 116 fills versus 306, with CAGR 45.72% versus 47.09% and maximum drawdown 43.28% versus 43.14%. It fails the training rule's 0.5-point growth tolerance in the holdout. The selected parameters are retained rather than revised after this result.

Research status. Retrospective analysis, not an investment recommendation or a verified trading record. Independent institutional-style presentation; no affiliation with an investment firm. Definitions, execution assumptions and limitations are documented inside.

01 / POLICY DEFINITION

1 Separate the trigger from the correction

A prior-close observation determines the target; the broker owns execution.

Let w_t be the AAPL holding value divided by total account equity at the close, with strategic target $w^* = 0.5$. Choose an absolute band h and correction fraction α:

$$\text{trigger at close } t \iff |w_t - 0.5| \geq h, \qquad \widehat{w}_{t+1} = w_t + \alpha(0.5 - w_t).$$

The TSLA target is $1 - \widehat{w}_{t+1}$. Initial investment is always 50/50, regardless of the trigger band. If the trigger does not fire, no target is sent and holdings drift. The observed AAPL weight uses total account equity, including residual cash; this is not a stock-sleeve-normalized weight.

Design choice	Specification
Band grid	0–10 percentage points, in increments of 0.25
Correction grid	10%, 20%, ..., 100% of prior-close deviation
Example	54% AAPL; 3.75 pp band; 20% correction: target 53.2%
Observation	Every session close; current close is available
Execution	Next exchange open; target held fixed through the gap
Baselines	Daily, weekly, monthly, quarterly; initial 50/50 buy and hold
Main period	Cash close 2020-01-03; buy 2020-01-06; value 2025-12-31
Chronological split	Train through 2022-12-30; fresh test entry 2023-01-03

What partial correction means

With a 20% correction, a sufficiently large deviation can remain outside the band after execution. It may therefore trigger again at the next close. The design intentionally allows repeated small corrections; it does not require a crossing from inside to outside the band. Trigger decisions, trade days and instrument fills are recorded separately because integer rounding can produce a signal with no executable quantity.

No promised weight cap

The threshold is a decision trigger, not a maximum weight deviation. Overnight gaps can change the opening allocation, partial correction leaves a residual deviation, and close-to-close movements can move the portfolio further before the next decision. The rule also does not create a stop-loss or cap drawdown.

The independent research harness retains the original broker's opening-equity sizing, adverse-price integer targets, sell-first execution and cash constraints. It does not modify the production QuantLab engine or add predictive factors.

02 / PARAMETER MAP

2 More tolerance buys less trading

Reduced turnover comes with looser adherence to the strategic allocation.

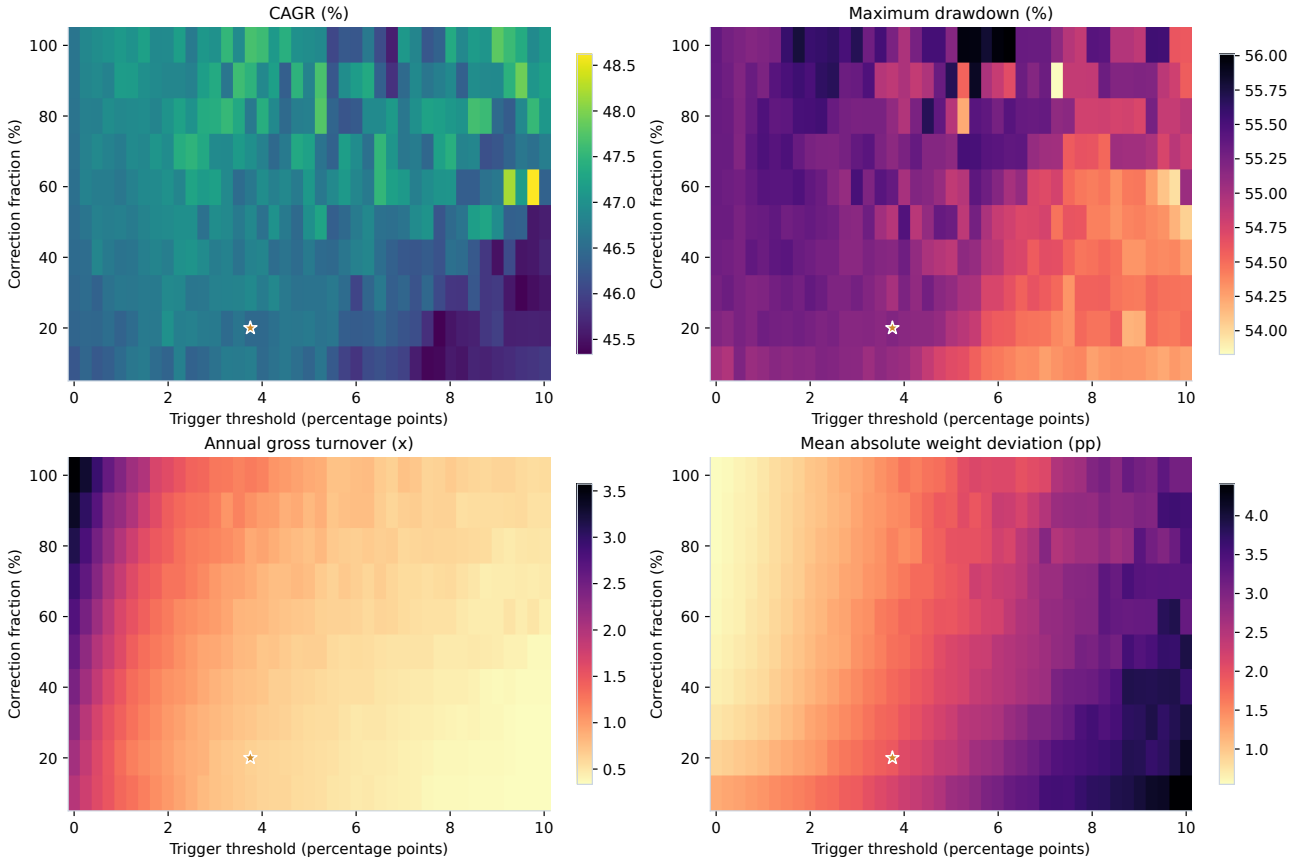


Figure 1: All 410 threshold policies under base costs. The star is the training-selected candidate, not the maximum-CAGR point. Threshold units are absolute percentage points; correction is a fraction of observed prior-close deviation. Each panel has its own scale.

The return and drawdown surfaces are irregular. There is no smooth improvement as the band widens or the correction shrinks. Calendar timing, retained relative trends and implementation costs all interact with the observed AAPL/TSLA path.

The clearest operational gradients are turnover and average deviation. Wider bands and smaller corrections generally permit more drift and fewer trades. At a zero band and 100% correction, the rule exactly reproduces the daily 50/50 reference, providing a useful limiting-case check.

A low-turnover policy can behave increasingly like buy and hold. Consequently, a return difference cannot be attributed to fee savings alone: the exposure path has also changed.

Read the operational panels before the return maximum. A policy that trades little but no longer tracks its intended allocation may solve a different portfolio problem.

03 / PRACTICAL TRADEOFF

3 Compare operating points, not only return peaks

The candidate saves trading while retaining substantially more control than buy and hold.

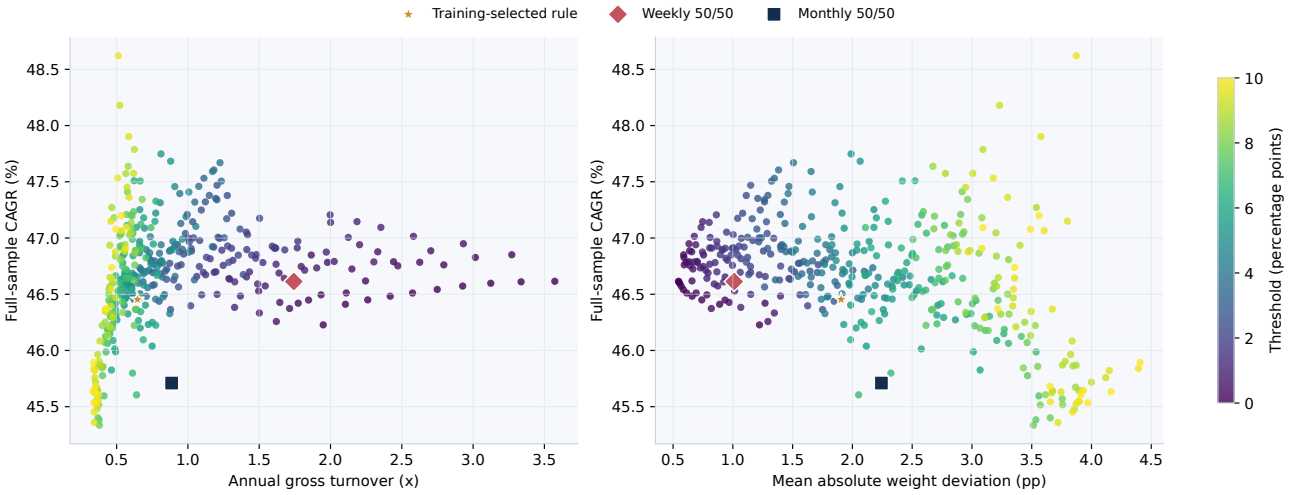


Figure 2: Full-sample growth versus turnover and average absolute closing weight deviation. Color indicates the threshold. The selected rule is determined by the training criterion; all full-sample scatter points are retrospective descriptions.

Full sample	CAGR	Max DD	Fills	Turnover	Mean deviation
Selected threshold	46.45%	55.20%	269	0.65x	1.91%
Daily	46.62%	55.36%	2884	3.57x	0.55%
Weekly	46.61%	55.31%	613	1.74x	1.01%
Monthly	45.71%	55.09%	140	0.89x	2.24%
Quarterly	45.26%	53.70%	46	0.57x	3.94%
Buy and hold	45.76%	66.41%	2	0.17x	25.98%

The selected threshold policy's mean absolute deviation from 50% is 1.91 percentage points, compared with 1.01 for weekly and 2.24 for monthly. Its turnover of 0.65 times opening equity per year is below both references. Its net growth is close to weekly in the full history, but that closeness is not imposed or assured in the later sample.

A dollar cost figure alone is insufficient for choosing a rule. Commission and slippage are lower for a policy that trades less, yet its changed exposures can either add to or subtract from returns. The research question is whether the complete policy is acceptable, not whether its fee line is smaller.

04 / PATHS AND RISK

4 Looser control is visible between trades

The target remains 50/50; the realized weight is allowed to wander.



Figure 3: Top: full-sample net wealth on a logarithmic scale. Bottom: observed AAPL closing weights for the selected, weekly and monthly rules. The buy-and-hold wealth reference is included above, while its much larger weight drift is omitted below for readability.

The candidate repeatedly allows moderate deviations and corrects them in small steps. Its weight path therefore differs from both a weekly reset and a monthly reset. The lower panel shows why fewer fills do not mean identical market exposure.

Full-sample maximum drawdown remains about 55% for both the candidate and the weekly reference. This policy changes trading behavior, not the strategic fact that the portfolio holds two volatile stocks. It cannot be characterized as a low-drawdown strategy merely because its turnover is low.

The first investment uses the original USD 100,000 capital scale. Integer adjusted units and minimum commissions matter at this scale. This paper does not vary account size, deposits, taxes or the target AAPL/TSLA ratio; those would be separate experiments.

05 / CHRONOLOGICAL CHECK

5 Freeze the rule before the later period

The candidate keeps its trading savings but misses the growth tolerance in the holdout.

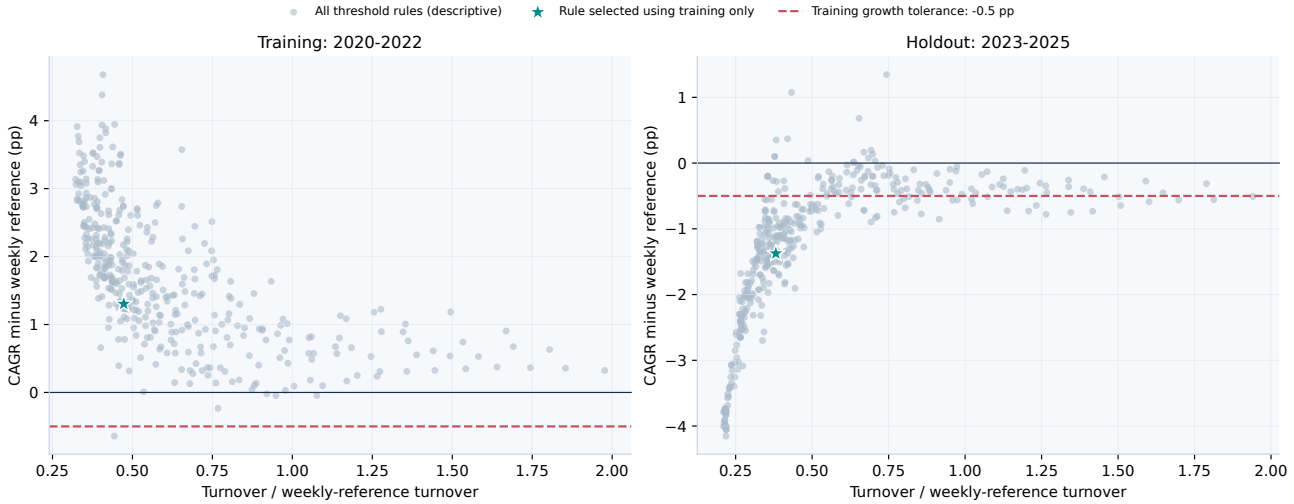


Figure 4: Each point is a threshold policy. The star is selected using training observations only. The dashed line is the training growth tolerance; the test cloud is descriptive and is not used to select a replacement.

The screening rule requires training CAGR at least weekly minus 0.5 percentage points, MDD no more than weekly plus two points, and mean absolute weight deviation no more than two points. Among feasible policies it minimizes annual gross turnover; ties prefer higher CAGR, smaller bands and larger corrections. There are 187 feasible training candidates. This is a retrospective research protocol saved before these calculations, not a preregistration made before the historical period.

Period / policy	CAGR	Max DD	Sharpe	Fills	Mean deviation
Training selected	48.24%	54.13%	1.055	161	1.95%
Training weekly	46.93%	54.21%	1.039	305	1.09%
Holdout selected	45.72%	43.28%	1.194	116	1.85%
Holdout weekly	47.09%	43.14%	1.219	306	0.94%

Fresh cash at 30 December 2022 close is invested at 3 January 2023 open; training holdings are not carried forward. In the later period, CAGR is 1.38 points below weekly, exceeding the allowed 0.5-point deficit. The drawdown and average-deviation tolerances remain satisfied. Thus the original composite acceptance condition does not pass on holdout data, despite substantial trading savings.

One split is limited evidence. The universe and question were chosen with knowledge of history, so the later period is an internal chronological diagnostic rather than an untouched external validation set. No formal statistical superiority or universal parameter optimum is claimed.

06 / COST STRESS

6 Fee savings are real but path dependent

Rerun holdings under changed costs rather than subtracting fees after the fact.

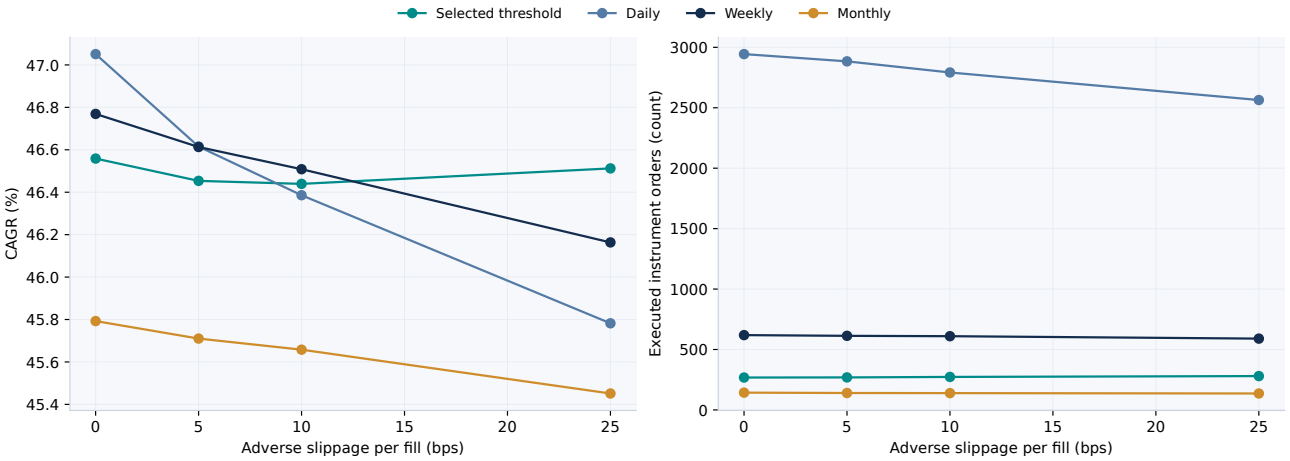


Figure 5: Free execution removes commission and slippage; nonzero cases retain the base commission and apply 5, 10 or 25 bps adverse slippage. Parameters remain fixed. Fill counts can change because costs alter affordability and integer quantities.

The fixed candidate is compared with daily, weekly and monthly policies under each cost setting. Every scenario recomputes quantities, cash and subsequent equity. The effect therefore includes compounding and changing exposures, rather than a simple retrospective deduction.

These costs are stylized. There is no estimation of opening-auction liquidity, spread variation during stress or market impact. Five basis points is the continuity assumption from the original study, not a measured execution guarantee.

Conclusion. Threshold and partial-correction rules offer an explicit way to exchange tighter tracking for less trading. This training-selected candidate preserves a large reduction in fills in 2023–2025 but gives up more growth than the research tolerance permits. The study supports the tradeoff, not a claim that the candidate dominates weekly rebalancing.

Evidence in context

Vanguard's published discussion of threshold rebalancing emphasizes the balance between transaction costs and adherence to strategic allocation. Its multiasset target-date implementation is motivation for this question, not evidence that the same thresholds work for AAPL/TSLA. The present policy is independently specified and evaluated on its own data.[1]

[1] Vanguard, *Vanguard's approach to target-date fund rebalancing*, https://workplace.vanguard.com/insights-and-research/perspective/vanguards-approach-to-target-date-fund-rebalancing.html.

07 / AUDIT

7 Reproduce the policy, not just the headline

Full numerical results and the reference execution code accompany this report.

Verification performed

The weekly reference reproduces the original archived closing equity path to a maximum difference of 3.50×10^{-10} USD. Six full-path comparisons against the frozen original broker include four new threshold policies, daily full correction and weekly rebalancing; all have zero observed equity difference in double precision. Zero band / full correction also matches daily exactly.

A future-close perturbation check confirms that changing closes from a specified session onward cannot affect opening targets through that session. Nonnegative cash and holdings and finite positive equity are asserted. These checks validate the research implementation under the stated convention; they do not validate the convention as a literal auction-order workflow.

Accounting and metrics

All historical opening-execution runs use USD 100,000 initial cash, two assets, no leverage or shorting, integer adjusted research units and nonnegative cash. Raw OHLC is scaled by adjusted close / raw close. Dividends and splits are not posted again. Adjusted units preserve the original QuantLab convention but are not historical legal share counts; per-unit commissions inherit this limitation.

Base costs are 5 bps adverse opening slippage and USD max(1, 0.005 times quantity) per executed instrument order. Sales precede buys; symbol order breaks purchase ties. Final holdings are marked without liquidation. Slippage is reflected in fills and is not deducted twice. No taxes, cash interest, capacity constraints or intraday drawdown estimates are included.

CAGR uses elapsed calendar days / 365.25. Annual volatility is the sample daily-return standard deviation times $\sqrt{252}$; Sharpe is the arithmetic mean divided by that standard deviation times $\sqrt{252}$, with zero risk-free rate. MDD is the positive magnitude of the largest decline from a running closing peak. Gross turnover sums opening-reference traded notional / pre-trade opening equity and divides by elapsed years, including entry; it is not divided by two. This turnover convention differs from the original engine's prior-close / fill-notional convention.

Files and reproduction

`data/` freezes market inputs and provenance. `results/` contains complete numerical tables, audit checks and compressed paths where applicable. `code/` contains calculation, plotting and report-generation scripts; `figures/` keeps vector PDFs and PNG previews. A SHA-256 manifest records delivered-file integrity. Run from the project directory in the documented Python environment:

```
python code/research.py
python code/validate.py
python code/plot_figures.py
python code/build_report.py
mkdir -p /tmp/quantlab-paper-build
pdflatex -output-directory=/tmp/quantlab-paper-build report.tex
pdflatex -output-directory=/tmp/quantlab-paper-build report.tex
```

Copy the compiled PDF from the temporary build directory. No network is required to rerun frozen-input calculations. The report builder combines computed tables with fixed English interpretations; changed assumptions require a review of those interpretations.

Input provenance and limitations

The unchanged AAPL/TSLA/SPY input is Yahoo Finance via yfinance 0.2.66, downloaded 18 September 2026. Only AAPL and TSLA enter these strategies. The original source SHA-256 is `9ea93af1777f51df9631178c2f029efcaf43958f35a9ba7538f8f173c91b9348`. Original reference equity, provider metadata, selection protocol, all grids and checks are retained. No new prices or synthetic histories enter this report.

The stock pair was selected retrospectively; a six-year sample and one split cannot establish robustness across assets or market cycles. Closing marks omit intraday losses. The partial-target rule intentionally differs from rebalancing to a band edge, and neither should be interpreted as a risk guarantee.